# O2 Desktop Workstation

Technical Specifications

## BASE SYSTEM HARDWARE FEATURES AND OPTIONS

**Processor**

R10000 @ 175MHz, 1MB secondary cache

R10000 @ 150MHz, 1MB secondary cache

R5000 @ 180MHz, 32KB data/32KB instruction primary cache, 512KBsecondary cache (on SC models only)

**Memory**

Synchronous DRAM, 100MHz, 4 banks, 288-bit wide

Capacity: 32MB–256MB with 16Mb SDRAM; 128MB–1GB with 64Mb SDRAM (available 1997)

**Mass Storage**

2 x Ultra Fast/Wide SCSI single-ended, 1 internal bus, 1 external bus (40MB/sec peak each)

2 x 1" x 3.5" internal drive bays (only 1 available on R10000 processor models)

Internal 4X CD-ROM

Hard drives: 2GB (5,400 RPM) and 4GB (7,200 RPM)

**Networking Hardware**

10BaseT/100BaseTX Ethernet standard (RJ-45)

PCI networking options: FDDI (dual-attach fiber, single-attach fiber, or unshielded twisted pair); 10BaseT/100BaseTX Ethernet; ISDN basic rate interface (S/T connector); Synchronous serial 2 x T1/E1; ATM**; Serial HIPPI**; Fibre Channel**

**I/O**

One half-length, 64-bit PCI slot (10W)

Optional digitaldisplay adapter card for Silicon Graphics® Presenter 1280 and Indy Presenter™

Additional SCSI channel available via PCI

**Input Devices**

PC (PS/2™) compatible keyboard and mouse

**Audio**

Analog stereo input/output

Powered speaker output

Monaural speaker

Microphone

**O2 Video I/O***

2 input channels and 1 output channel

S-Video and composite video I/O (S-Video with $I^2C$ control)

Digital camera

CCIR601 digital video**

**Monitor**

20" (19" viewable) Trinitron® 1280x1024

17" (16" viewable) Trinitron 1280x1024

**Connectors and Controls**

Front: Volume control (up/down); Speaker; Status LED; Reset switch; Power on

Rear: 2 serial 460kbps (DB-9); Monitor (HD15) PC style; Ultra Fast/Wide SCSI (H-den 68); 10BaseT/100BaseTX Ethernet (RJ45); 1284 parallel (C miniature); Mouse/keyboard (2 x MDIN6)

Side: Right and left audio line in/out (RCA); Stereo headphone jack (3.5mm); Microphone jack (3.5mm); S-Video I/O (MDIN)*; Composite video I/O (RCA)*

**Rear Audio/Video**

Digital video I/O and camera (68-pin)*

Right and left audio line I/O powered (2.5mm power, 3.5mm audio)

**Package**

9" W x 12" H x 10.5" D

22 lbs

175-watt power supply

## GRAPHICS

**Resolution**

1280x1024 @ 75Hz

1024x768,800x600, 640x480 also supported

**Frame Buffer Formats**

32 + 32-bit FB (32 + 32 double buffered)

32-bit FB (16 + 16 double buffered)

16-bit FB (8 + 8 double buffered)

8-bit overlay

**Graphics Features**

Native OpenGL graphics subsystem

Hardware z-buffer

Triangle rasterization in hardware

Texture mapping in hardware

Hardware image mapping support

Hardware stencil planes

Hardware anti-aliasing

Source plus destination alpha in hardware

Fast Xline performance

## VIDEO COMPRESSION

**Hardware accelerated**

– JPEG encode/decode
30fps @NTSC or CCIR 601 resolution
25fps @PAL or CCIR 601 resolution

**Software based**

– H.261 encode/decode
– QuickTime encode/decode
– Cinepak encode/decode
– AVI encode/decode
– MPEG-1 encode/decode

## STANDARD SOFTWARE FEATURES

**Operating System**

IRIX™ 6.3 with XFS™

**Networking**

TCP/IP, Novell NetWare™ support, Xinet AppleTalk® support,Windows® networking, ONC with NFS 3.0 (optional)

**Run-Time Libraries**

OpenGL and OpenGL Imaging Extensions

ImageVision Library®

Digital media libraries

## REMOVABLE MEDIA OPTIONS

Floppy (parallel port), 4.0GB DAT, SyQuest®**,10GB or 15GB digital linear tape, 1.3GB optical disk**, ZIP, JAZ,external 3-bay enclosure

## INPUT DEVICE OPTIONS

Optical mouse**, Magellan**,Spaceball**, dials and buttons,tablets**

## MISCELLANEOUS OPTIONS**

Macintosh® MIDI port converter

RS-422 port converter

## REGULATORY COMPLIANCE

FCC/CISPR 22 class B

IEC 950 CSA/TUV

EV EMC compliant

## SUPPORT

Warranty: 1 year advanced parts exchange; MTBF 70,000 hours (est.)

*not present on all models

**not sold by Silicon Graphics

| Performance | R5000/ 180MHz PC | R5000/ 180MHz SC | R10000/ 150MHz SC | R10000/ 175MHz SC |
|---|---|---|---|---|
| **Viewperf 5.0 (frames/sec)** | | | | |
| CDRS Viewset Composite | 11.4 | 12.24 | 11.52 | 13.57 |
| Data Explorer Viewset Composite | 2.38 | 2.47 | 2.14 | 2.46 |
| Design Review Viewset Composite | 1.99 | 2.08 | 1.79 | 2.01 |
| Advanced Visualizer | 4.17 | 4.41 | 4.18 | 4.91 |
| Lightscape | 0.40 | 0.41 | 0.35 | 0.41 |
| Xlines | 2.5M | 3.2M | TBA | TBA |
| Triangles (Lit, Gouraud,Z, 50 pixel) | 377 K/sec | 377 K/sec | 330 K/sec | 382 K/sec |
| Bilinear texture fill rate | 40 MP/sec | 40 MP/sec | 40 MP/sec | 40 MP/sec |
| Trilinear texture fill rate | 30 MP/sec | 30 MP/sec | 30 MP/sec | 30 MP/sec |
| Pixel fill rate | 65 MP/sec | 65 MP/sec | 65 MP/sec | 65 MP/sec |
| SPEC int 95 | 3.4 | 4.8 | 6.6 | 7.6 |
| SPEC fp | 4.5 | 5.4 | 6.2 | 6.6 |
| AIM | 208 | 242 | 353 | 426 |

*TBA–To Be Announced*


SiliconGraphics Computer Systems

**Corporate Office**
2011 N. Shoreline Boulevard
Mountain View, CA 94043
(415) 960-1980
**URL:http://www.sgi.com**

U.S. 1(800) 800-7441
Europe (44)1 734-25.75.00
Asia Pacific (81) 3-54.88.18.11
Intercontinental 1(415) 933.46.21
Latin America 1(415) 933.46.37
Canada 1(905) 625-4747
Australia/New Zealand (61) 2.9879.9500



© 1996 Silicon Graphics,Inc. All rights reserved.Specifications subject to change without notice. Silicon Graphics,OpenGL, ImageVision Library, Indigo, and the Silicon Graphics logo are registered trademarks, and O2, XFS,Indigo2, Indigo2 IMPACT, InfiniteReality, WebMagic, InPerson, and IRIX are trademarks,of Silicon Graphics,Inc. Indy is a registered trademark,and Indy Presenter is a trademark,used under license in the U.S.and owned by Silicon Graphics,Inc.,in other countries worldwide. MIPS and the MIPS RISC Certified Power logo are registered trademarks,and R5000 and R10000 are trademarks,of MIPS Technologies,Inc. Ethernet is a trademark of Xerox Corporation.NFS is a trademark of Sun Microsystems,Inc. SyQuest is a registered trademark of SyQuest Technologies,Inc . Macintosh,Cinepak,AppleTalk,and QuickTime are registered trademarks of Apple Computer, Inc. PS/2 is a trademark of IBMCorporation.NetWare is a trademark of Novell,Inc. Windows is a registered trademark of Microsoft Corporation. Trinitron is a registered trademark of Sony Corporation. Netscape and Netscape Navigator are trademarks of Netscape Communications Corporation. UNIX is a registered trademark in the U.S.and other countries,licensed exclusively through X/Open Company Limited.SoftWindows is a trademark of Insignia Solutions,used under license. Motif is a trademark of Open Software Foundation. PostScript is a registered trademark of Adobe Systems,Inc. All other trademarks mentioned herein are the property of their respective owners.Image Credits: Cover: Underwater Camera,courtesy of Katz Design,Inc.,Montreal,Canada. *Page 2-3:* Mask,courtesy ofThe MacNeal-Schwendler Corporation,Cellular Phone, courtesy of Katz Design,Inc.,Montreal,Canada. *Page 6,from top to bottom:* Zebra,courtesy of Christine Low; IBM Atrium, © 1995 A.J.Diamond, Donald Schmitt & Co. (Toronto, Canada), Image created with the Lightscape Visualization System; Critic.com,courtesy of Kronos Digital Entertainment,Inc., Electron density contour plot of porphyrin,courtesy of Wavefunction,Inc. Contour Zone, image created courtesy of William M.Brown of the U.S.Army Construction Engineering Research Labs using GRASS GIS;Saw, © 1995,1996 Structural Dynamics Research Corporation. *Page 8,from top to bottom:* Heads Will Roll,Wayne Bretcha, Animator/Designer and Ed Newman,Animation Director; Reboot,courtesy of Mainframe Entertainment,Inc.,Video screen shot,© 1995,1996 Structural Dynamics Research Corporation;Cellular Phone screen shot and Underwater Camera screen shot,courtesy of Katz Design,Inc.Montreal,Canada.


(1/97)

# O2™ Desktop Workstation



Product Guide

COMBINING WORKSTATION PERFORMANCE WITH THE POWER OF THE WEB

# Performance

The O2 desktop workstation is the innovative system that combines Silicon Graphics performance with the power of a Web-integrated user environment. O2 gives you the industry-leading CPU and graphics performance you need plus breakthrough video and imaging features you never expected to have, especially at such an affordable price. With this combination of performance and outstanding Web capabilities, you'll accomplish more on the desktop than you ever imagined possible.



**O2 STANDARD FEATURES**

- MIPS® R10000™ or R5000™ processor
- IRIX™ 6.3 operating system
- Web-integrated user environment
- 32-bit double-buffered graphics
- Hardware-accelerated texture mapping
- Hardware-accelerated z-buffer
- Image processing engine
- Video compression engine
- 64MB system RAM*
- 2GB system disk
- 10BaseT/100BaseTX Ethernet™ networking
- Two Ultra Fast/Wide SCSI buses
- Stereo audio
- 64-bit PCI expansion slot
- CD-ROM
- 17" monitor (1280x1024)

** Entry configuration has 32MB/1GB*

### Powering Your Organization

O2 is designed for professionals on the critical path—the creative and engineering teams that drive a company's competitive edge in manufacturing, entertainment, the sciences, and other demanding industries. Merging leading compute performance with the industry's first high-performance Unified Memory Architecture (UMA), O2 delivers incredible internal bandwidth to accelerate graphics and compute-intensive applications. This architecture has allowed Silicon Graphics to make this level of performance available at an affordable price.

### Interactive Graphics, Video, and Image Processing

The O2 Unified Memory Architecture comes standard with stunning 3D graphics, powerful image processing, and real-time video processing far beyond any other machine available in its class. These features make O2 the ideal workstation for CAD modeling, post-production, Web site creation, and scientific visualization. Standard compression hardware and software transforms any user into a video editor, while providing the quality that video professionals demand.

### Complete Web Integration

O2 combines the performanceof a SiliconGraphics® workstation with the power of the World Wide Web and is the first system to provide bundled professional digital media capabilities and Web authoring tools. With the Web integrated throughout the interface, users communicate their work easier than ever before on company intranets and the Internet.



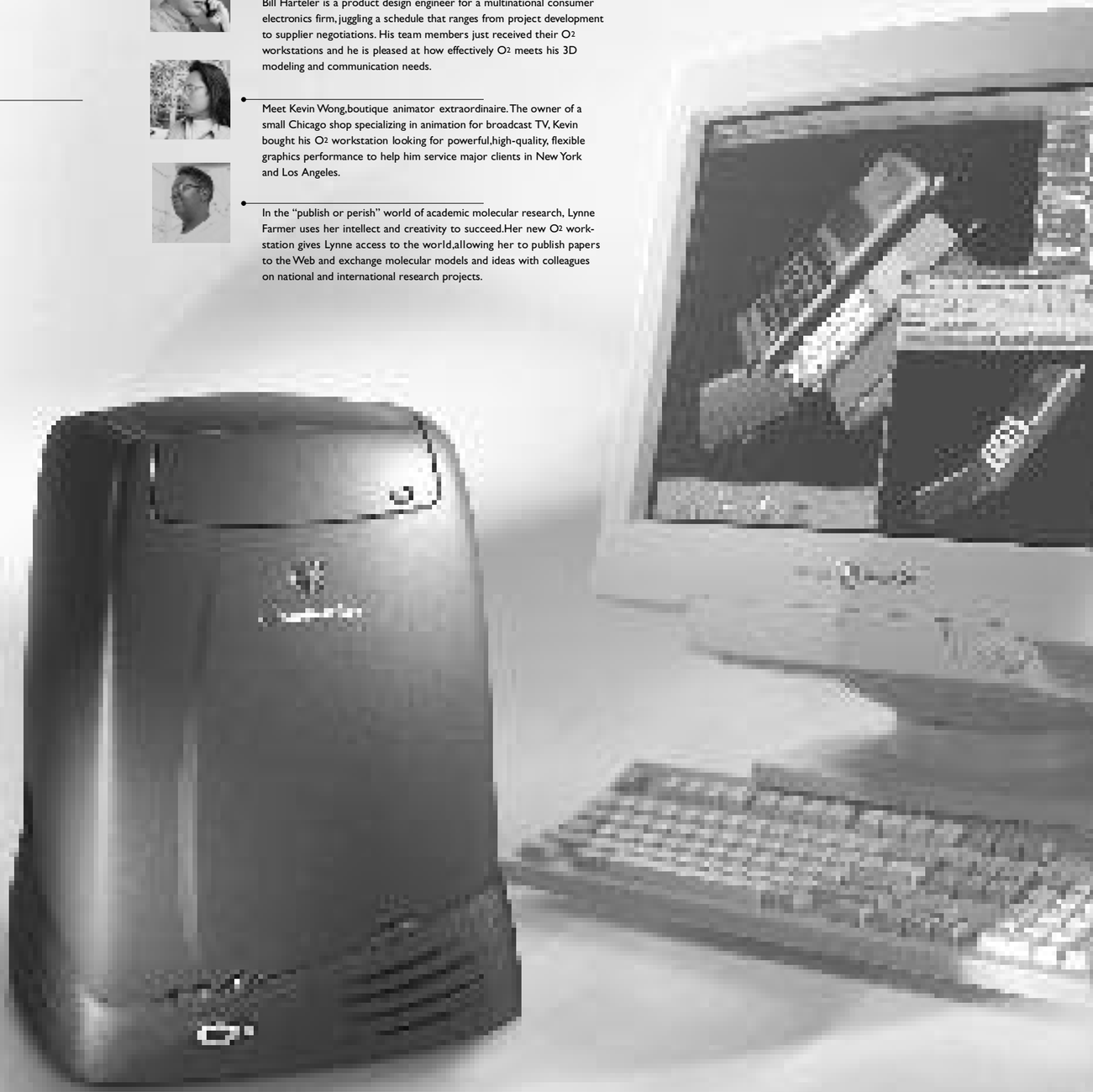
Bill Harteler is a product design engineer for a multinational consumer electronics firm, juggling a schedule that ranges from project development to supplier negotiations. His team members just received their O2 workstations and he is pleased at how effectively O2 meets his 3D modeling and communication needs.



Meet Kevin Wong, boutique animator extraordinaire. The owner of a small Chicago shop specializing in animation for broadcast TV, Kevin bought his O2 workstation looking for powerful, high-quality, flexible graphics performance to help him service major clients in New York and Los Angeles.



In the "publish or perish" world of academic molecular research, Lynne Farmer uses her intellect and creativity to succeed. Her new O2 workstation gives Lynne access to the world, allowing her to publish papers to the Web and exchange molecular models and ideas with colleagues on national and international research projects.



# Powerful

With the O2 desktop workstation, Silicon Graphics has created the industry's first high-performance Unified Memory Architecture. UMA transcends inflexible personal computer and traditional workstation designs,which constrain performance with distributed graphics,imaging, and video memory. O2 delivers extraordinary performance by integrating powerful compute resources—video, compression, graphics, image processing,and I/O—via a high-bandwidth,low-latency memory system.



**KEY CPU FEATURES**

**R5000 CPU**
- MIPS IV instruction set
- 2-way superscalar, 64-bit architecture
- 32KB instruction cache
- 32KB primary cache
- 512KB secondary cache *available in SC configuration*

**R10000 CPU**
- MIPS IV instruction set
- 4-way superscalar, 64-bit architecture
- Out-of-order instruction execution
- 32KB instruction cache
- 32KB primary cache
- 128-bit dedicated secondary cache bus
- 5 separate execution units
- 1MB fast secondary cache

**KEY ARCHITECTURE FEATURES**
- 2.1GB memory bandwidth
- 256MB memory (1GB available in 1997)
- Memory subsystem uses commodity synchronous DRAMS
- ECC memory protection for high memory integrity

### High-Performance Unified Memory Architecture

Unlike typical PC and workstation architectures, where data must be transferred between graphics, video, and imaging memory located on separate boards, all O2 data resides in main memory where every engine has direct access to it. With UMA, system memory, frame buffer, z-buffer, texture memory, rendering memory, image memory, and video memory are all the same. No longer do dedicated pools of memory exist throughout the computer to be utilized only at specific times; with O2, memory is automatically reallocated according to need. Expanding texture memory becomes a matter of simply buying a standard memory upgrade.

### High-Bandwidth I/O

The O2 I/O engine maximizes performance by removing bandwidth bottlenecks, which can starve even the fastest system components. O2 delivers peak performance on 10BaseT/100BaseTX Ethernet networks, a dual Ultra Fast/Wide SCSI implementation, a 64-bit PCI expansion bus, and several other standard I/O options.

### Leading Processing Power

The O2 workstation is powered by either the 64-bit MIPS R5000or R10000 processors,delivering industry-leading application performance. With the R5000 processor, users get atrue price/performance leader, a CPU that delivers excellent compute, graphics, and application performance. The R10000 processor provides leading compute power for the most demanding applications.

### A Reliable Modular Design

The elegant O2 system has a five-piece modular design to simplify upgrades and maintenance—disk drives, system module, and PCI cards can be easily accessed from the rear of the system. This simplicity is also reflected in the O2 system administration tools, which guide users through simple maintenance and configuration functions. Further servicing is available through a series of warranty options and on-line support systems.



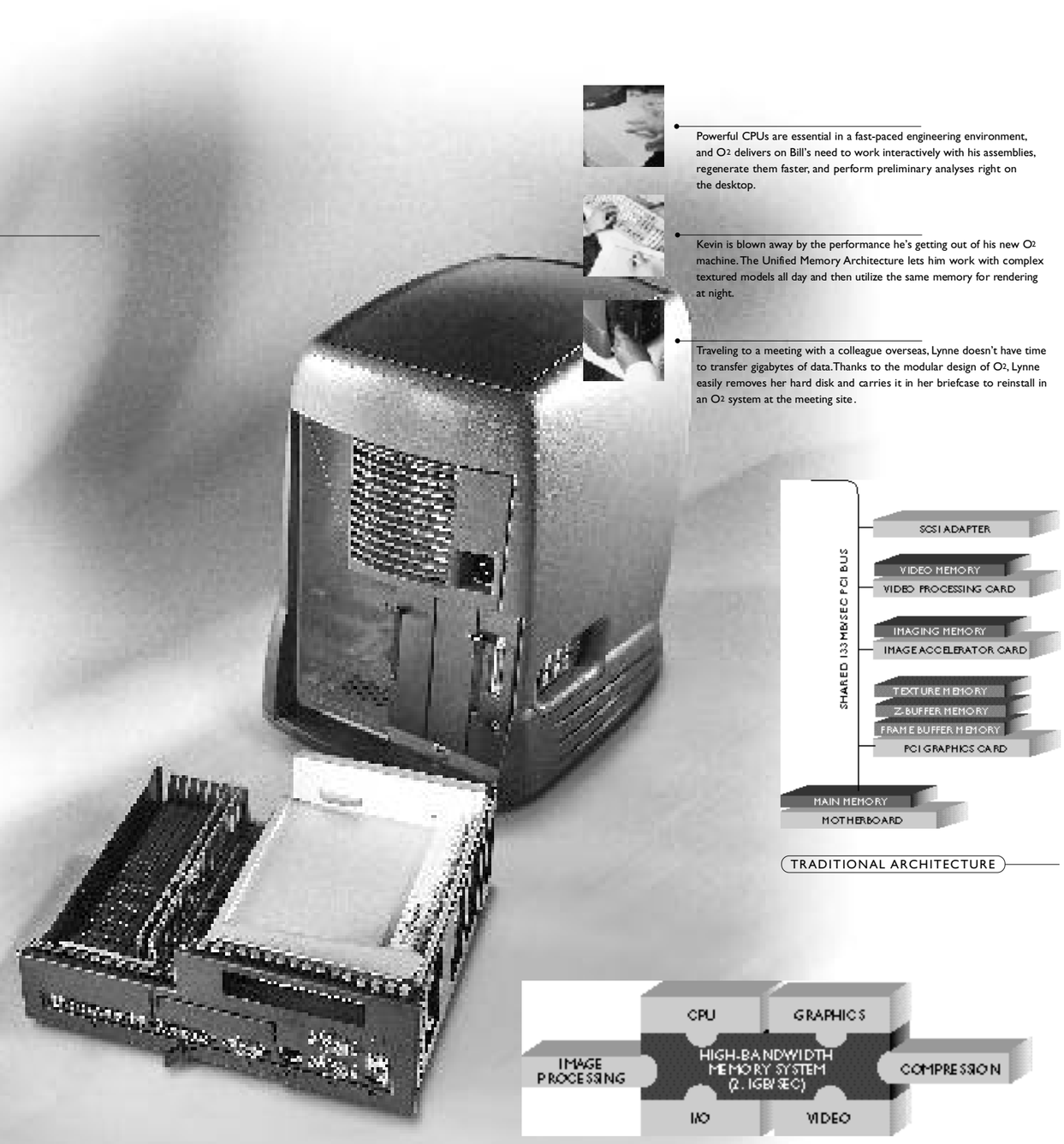
Powerful CPUs are essential in a fast-paced engineering environment, and O2 delivers on Bill's need to work interactively with his assemblies, regenerate them faster, and perform preliminary analyses right on the desktop.



Kevin is blown away by the performance he's getting out of his new O2 machine. The Unified Memory Architecture lets him work with complex textured models all day and then utilize the same memory for rendering at night.



Traveling to a meeting with a colleague overseas, Lynne doesn't have time to transfer gigabytes of data.Thanks to the modular design of O2, Lynne easily removes her hard disk and carries it in her briefcase to reinstall in an O2 system at the meeting site.




TRADITIONAL ARCHITECTURE




O2 UNIFIED MEMORY ARCHITECTURE OVERVIEW


www.sgi.com/O2 WEB

# Interactive

The O2 desktop workstation's industry-leading 3D graphics and image processing embody everything you've ever heard about Silicon Graphics performance and realism. Built upon a native OpenGL® graphics subsystem and Unified Memory Architecture, O2 delivers a level of interactivity not available with any other machine in its class. These remarkable capabilities are standard with every configuration, giving you all the graphics and image-processing performance you need, right out of the box.



**GRAPHICS FEATURES**

- 1280x1024 at 75Hz (also supports VGA, SVGA, XGA)
- Up to 32-bit RGBA double-buffered standard
- Native OpenGL graphics subsystem
- Hardware z-buffer
- Triangle rasterization in hardware
- Texture mapping in hardware
- Hardware image mapping support
- Hardware stencil planes
- Hardware anti-aliasing
- Source plus destination alpha in hardware
- Fast Xline performance

*For applications requiring additional screen space, the O2 workstation supports a dual-head option through a third-party PCI card.*

### Advanced Graphics Features

With standard 32-bit double-buffered graphics and advanced features accelerated in hardware, O2 brings power and high quality within the reach of every user. Accelerated features include texture mapping, z-buffer, and anti-aliased points and lines, as well as stencil, fog, and color space conversions.

### High-Performance Texture Mapping

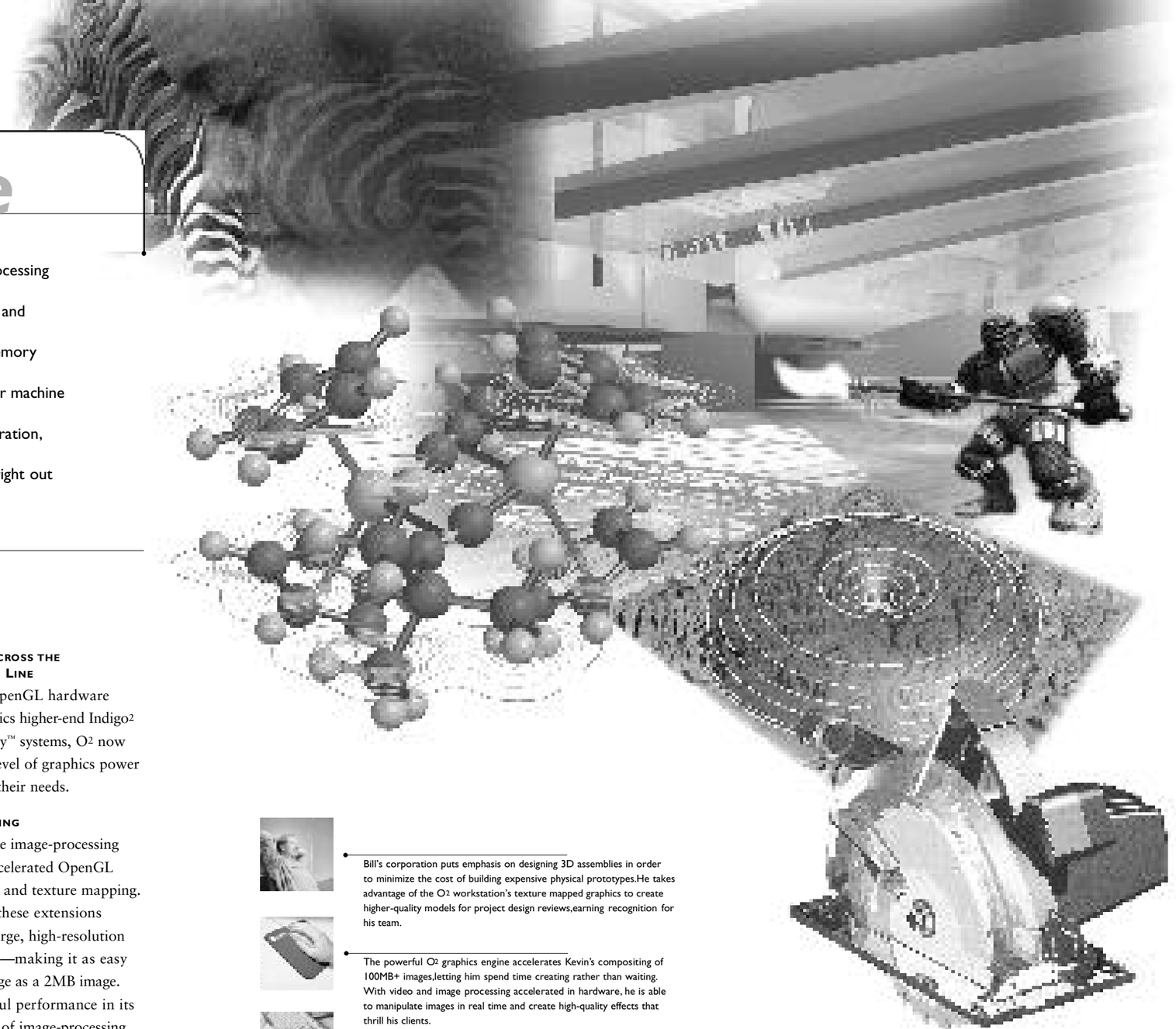
The O2 workstation's hardware-accelerated texture mapping capabilities bring a new level of realism and interactivity to the desktop. In CAD and animation, texture mapping allows users to visualize their models with a level of realism that surpasses traditional shaded models. Unlike traditional graphics boards that set a limit on texture memory, the flexible Unified Memory Architecture allows an unlimited amount of memory to be allocated for textures.

### Consistent Feature Set across the Silicon Graphics Product Line

Implementing all the key OpenGL hardware features of the Silicon Graphics higher-end Indigo2 IMPACT™ and InfiniteReality™ systems, O2 now allows users to choose the level of graphics power at the price that best meets their needs.

### First-Class Image Processing

O2 delivers high-performance image-processing capabilities via hardware-accelerated OpenGL image-processing extensions and texture mapping. Implemented in hardware, these extensions allow users to manipulate large, high-resolution image data sets in real time—making it as easy to manipulate a 200MB image as a 2MB image. Delivering the most powerful performance in its class, O2 raises the standard of image-processing power in a desktop workstation.





Bill's corporation puts emphasis on designing 3D assemblies in order to minimize the cost of building expensive physical prototypes. He takes advantage of the O2 workstation's texture mapped graphics to create higher-quality models for project design reviews, earning recognition for his team.



The powerful O2 graphics engine accelerates Kevin's compositing of 100MB+ images, letting him spend time creating rather than waiting. With video and image processing accelerated in hardware, he is able to manipulate images in real time and create high-quality effects that thrill his clients.



Instead of working on dedicated lab machines, Lynne finds that her research is enhanced by the O2 workstation's photo-realistic representation of molecular models right on her desktop, allowing her to more accurately analyze data and achieve new research insights faster.



www.sgi.com/O2 WEB

THE FIRST NATIVE VIDEO WORKSTATION

# Flexible

The O2 desktop workstation is truly a native video machine—the first system to integrate video, audio, and real-time compression technologies as fundamental components of its architecture. The flexible O2 architecture allows video to be brought directly into memory as a standard data type. Once there, the graphics, image-processing, and compute engines can access and manipulate the data in real time.





**STANDARD VIDEO FEATURES**

- Real-time record to disk using two independent SCSI channels
- Selectable genlock source: video input,internal clock, external,specified source
- Storing uncompressed video to disk using one SCSI channel; requires external disk capable of storage
- Synchronization of audio and video;field and sample accurate
- Real-time, JPEG compression/ decompression
- Bundled video control and software
- Alpha channel output support for switching and keying
- Real-time video effects using GL,video, and mip-map generation
- Two-way conversion, non-square to square pixel, for I/O
- Real-time color space conversion
- Dual input video stream DMA

**STANDARD AUDIO FEATURES**

- Analog audio sample rates: 8KHz-48KHz.
- High-quality 78dB signal-to-noise ratio
- Audio and video clock sync and sample accurate time stamping

### Flexible Video Processing

With every engine able to access all data residing in main memory, O2 delivers video manipulation capabilities never before available in this class. Applications can decode a compressed video source and use it as a texture map, or utilize the image-processing hardware to blur or distort a live video stream in real time. Users view video in its native format due to the O2 workstation's ability to display non-square video pixels.

### Professional Video Capabilities

O2 is the first desktop workstation to include real-time JPEG compression and decompression hardware in every system. Supporting compression ratios of up to 4:1, O2 delivers a level of quality that meets the needs of the nonlinear video editing marketplace. Independent audio can be synchronized to video data. O2 provides two channels of simultaneous input and one channel of output for D1 and analog video.

### Cross-Format Video Output

In addition to real-time capabilities, O2 implements a wide range of video compression algorithms through software, including industry standards such as QuickTime®, AVI, and Cinepak®. These built-in capabilities allow users to create and edit video on O2 and then distribute it via the Web to any computer for playback.

### Capturing the Screen Display as Video

O2 turns your application into a video source by allowing any portion of the screen to be recorded in real time directly to disk, negating the need for a VCR to enable video editing. Users can also output the screen recording to external video via the composite video or S-Video ports.



Due to a last-minute design change, Bill's tool vendor in Singapore needs to see its impact on the project. Within minutes,Bill captures a video sequence from his CAD application, edits the movie file using the O2 workstation's bundled media tools,and publishes it to his Web page for review in Singapore.



O2 lets Kevin create video effects on his desktop that only the big shops in New York used to be able to do.With video texturing and the ability to create his own 3D transitions and effects, Kevin has made his small business more valuable to his clients.



An important part of Lynne's job is sharing research results. Using bundled digital media tools, O2 lets her create QuickTime movies of complex molecular models with titles, audio, and annotations. She can then drag-and-drop the results to her Web site for review, impressing her colleagues.



www.sgi.com/O2 WEB

# Productive

The O2 desktop workstation provides users with a complete, easy-to-use environment for accelerating workflow and enhancing productivity. Extending the desktop to merge it with the Web, O2 offers users a total solution for creating,browsing,and serving data on intranet and Internet networks. An unparalleled set of Web and digital media tools allows you to communicate with colleagues and clients in more powerful ways than ever before.



**O2 FEATURES**

**IRIX 6.3**

*Built on UNIX® SVR4 with Berkeley Extensions 4.3,IRIX conforms to every major UNIX standard as well as a variety of cross-platform standards:*

- Motif™ 1.2.4
- X11R6
- POSIX 1003.1/1003.2
- FIPS 151.2
- Display PostScript®
- Tooltalk
- Triteal CDE (through Triteal)
- AppleTalk
- XFS™
- NFS™
- QuickTime
- Cinepak
- JPEG
- MPEG 1
- AVI file formats

### Built-In Intranet Support

Every O2 ships with an intranet-enabling tool suite that includes Netscape Navigator™ 3.0, Netscape Mail, and Netscape News Reader 2.0, as well as personal Web publishing software from Silicon Graphics, which utilizes Netscape FastTrack server and WebMagic™ Pro authoring tools. Web browsing and TCP/IP network protocols are completely integrated into the desktop user interface and operating system.

### Professional-Quality Media Tools

Outstanding bundled digital media tools give any user the ability to easily develop compelling digital media content for design reviews and project Web pages that incorporate video, audio, and 3D graphics.

### Insignia's SoftWindows™ 95

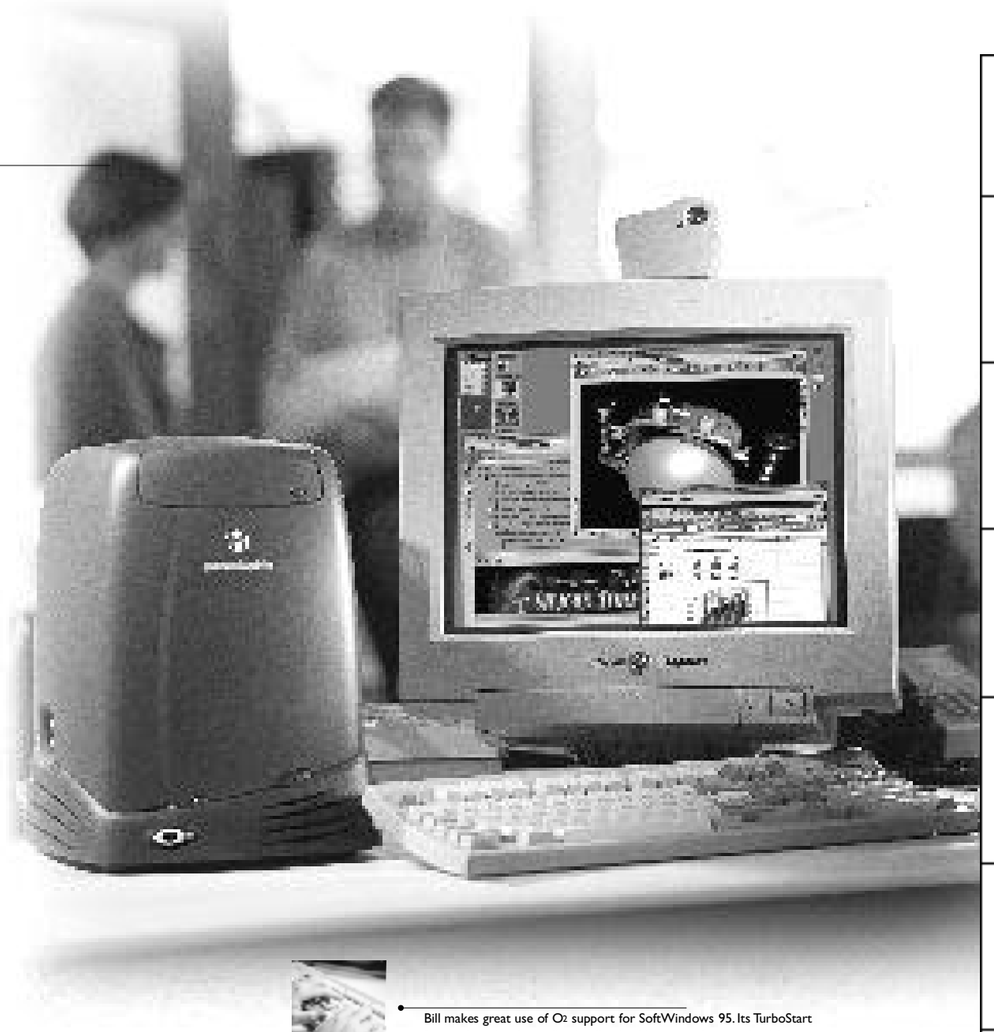
O2 supports Insignia's SoftWindows 95 to give users access to all the most popular PC applications. SoftWindows 95 leverages O2 hardware to accelerate multimedia playback. O2 also provides seamless integration into NetWare™ and AppleTalk® networks.

### Web-Based Help and System Administration Tools

Web-based help lets O2 users access documentation anywhere on their systems, networks, or the Internet, ensuring access to the most up-to-date technical information. The task-based system administration utilizes the intuitive look-and-feel of the Web and guides users through all system management procedures.

### Complete Real-Time Collaboration Solution

The Silicon Graphics InPerson™ 2.2 videoconferencing tool provides real-time collaboration that spans any geographical distance. It includes a 2D and 3D whiteboard that allows users to view, manipulate, and attach digital media annotations and communicate via the O2 digital camera.





Bill makes great use of O2 support for SoftWindows 95. Its TurboStart feature launches immediately, allowing him to run key productivity applications as well as the same scheduling program he has at home.



As a boutique shop owner, Kevin is also his own marketing department. Using the O2 workstation's bundled Web authoring tools,he's created a compelling presence on the Web that makes him look like one of the animation industry's major players.



Lynne makes heavy use of the drag-and-drop, network-aware user interface, which lets her access FTP and Web sites that she uses everyday in her research.

- Netscape Navigator 3.0 for Web browsing, e-mail,and news
- Personal Web publisher
- Netscape FastTrack server for Web serving solutions
- Internet-enabled file manager
- Web-based system administration and set-up tools
- Professional-quality digital media tools
- Bundled real-time conferencing and multimedia presentation tools



www.sgi.com/O2 WEB